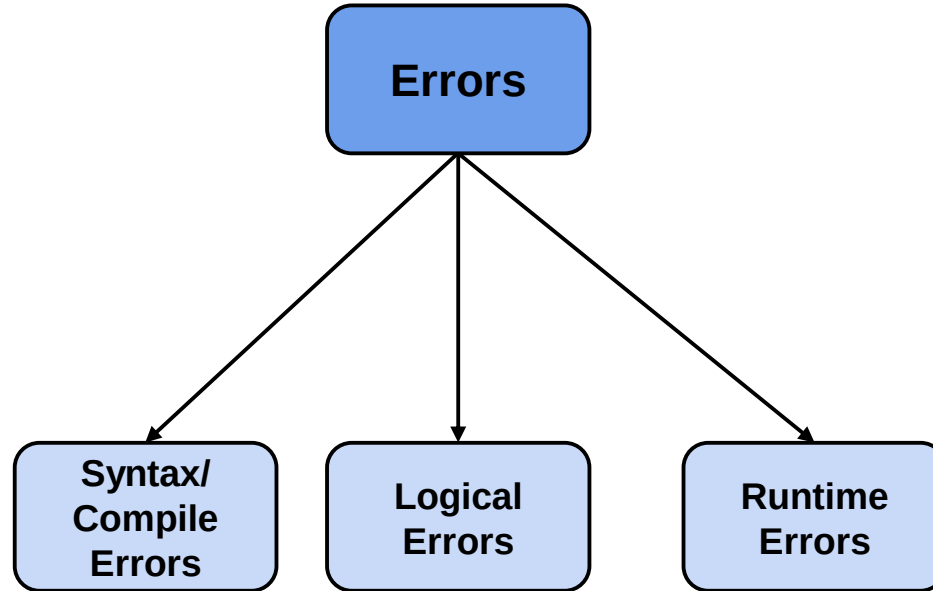


Understanding Errors

What is an error?

- A problem or exception that occurs during the execution of a program.
- Prevents the program from functioning correctly.
- Remains undetected until the program is compiled or run.

Types of Errors



Syntax Errors

- Occurs when the code violates the rules of the Java programming language's syntax.
- Also known as compile-time error.
- **Consequence:** The program cannot be successfully compiled and needs to be fixed before it can be executed.
- Easy to identify because the Compiler explicitly mentions the error when compiling the code.

Syntax Errors (Continued)

```
1. class HelloWorld {  
2.     public static void main (String[] args) {  
3.         System.out.println("Hello, World!");  
4.     }  
5. }
```

1 error found:

File:C:\Users\Desktop\SyntaxError.java[line:3]

Error: ')' expected

Syntax Errors (Continued)

```
1. class SyntaxError {  
2.     public static void main (String[] args) {  
3.         int x = 10  
4.     }  
5. }
```

1 error found:

File:C:\Users\Desktop\SyntaxError.java[line:3]

Error: ';' expected

Syntax Errors (Continued)

```
1. class SyntaxErrorExample2 {  
2.     public static void main (String[] args) {  
3.         System.out.println(value);  
4.     }  
5. }
```

1 error found:

File:C:\Users\Desktop\SyntaxErrorExample2.java[line:3]

Error: cannot find symbol

symbol: variable value

location: class SyntaxErrorExample2

Logical Error

- Refers to a mistake in the program's logic that leads to incorrect results.
- Can't detect during runtime or compile time.
- Hardest to identify.

Logical Errors (Continued)

```
1. class LogicalError {  
2.     public static void main (String[] args) {  
3.         int num1 = 10, num2 = 20;  
4.         System.out.println("Summation = " + (num1-num2));  
5.     }  
6. }
```

Welcome to DrJava. Working directory is C:\Users\Desktop

> run LogicalError

Summation = -10

This program won't show any error but it is logically incorrect because the output should be 30.

Runtime Errors

- Errors that occur while running the program.
- Not caught during Compilation time.
- Quite common type of errors but easily traceable.
- The Runtime Errors are actually different Exceptions being caught.

N.B. You'll learn more about Exceptions in the later courses.

Runtime Errors (Continued)

```
1. class RuntimeError {  
2.     public static void main (String[] args) {  
3.         int num1 = 10;  
4.         System.out.println("Division = " + (num1/0) );  
5.     }  
6. }
```

Welcome to DrJava. Working directory is /home/iqra/Documents

> run `RuntimeError`

`java.lang.ArithmeticException: / by zero`

This program won't know there's an error while compiling but when it tries to execute it'll realize that an integer is being divided by 0 and will throw **ArithmeticException**.

THANK YOU!